

> "Every Byte of Evidence Matters. Every Action is Traced."

SDIS

SECURE DIGITAL INVESTIGATION SYSTEM

A Secure Digital Case & Evidence Management Platform

- SECURE ACCESS CONTROL
- DIGITAL CASE MANAGEMENT
- EVIDENCE INTEGRITY
- AUDIT LOGGING
- TAMPER DETECTION

DEVELOPED IN
C++17 (Standard Library)

SYSTEM TYPE
Command-Line Application

PROJECT CATEGORY
Cybersecurity & Digital Forensics

USER ROLES
Detective | Commander

YEAR
2026

EVENT
Hackathon Project

SECURING EVIDENCE.
ENSURING INTEGRITY.
DELIVERING JUSTICE.

SECURE DIGITAL

INVESTIGATION SYSTEM

SDIS

A secure, command-line case and evidence management platform built for digital forensics and law enforcement operations. Every byte of evidence is tracked. Every action is traced.

PLATFORM	LANGUAGE	USER ROLES	YEAR
Command-Line (CLI)	**C++17 Standard Library**	**Detective & Commander**	**2026 — Hackathon**

1. WHAT IS SDIS?

The Secure Digital Investigation System (SDIS) is a fully self-contained case and evidence management platform designed around the principles of real-world digital forensics. It gives law enforcement personnel a structured, auditable environment to create cases, log evidence, track suspects, and monitor every action taken within the system.

Built from scratch using only C++17's standard library — no third-party frameworks, no external databases — SDIS demonstrates that strong security principles can be embedded directly into software architecture. Every feature, from login to tamper detection, is implemented intentionally.

The system is guided by one core principle:

> ***"Every byte of evidence matters. Every action is traced."***

2. WHY IT EXISTS

In real digital forensics and law enforcement, two things are legally critical: evidence integrity and chain of custody. If evidence cannot be proven untampered, or if the trail of who accessed what is missing, a case can collapse entirely in court.

SDIS was built to demonstrate how these principles — integrity verification, access control, audit logging, and tamper detection — can be baked into software from the ground up, not bolted on as an afterthought.

Real-world problems SDIS addresses:

- Unauthorized access to sensitive case data
- Undetected modification or tampering of evidence
- No record of who accessed or changed information
- Investigators viewing cases they should not have access to
- No oversight mechanism for supervisory staff (Commanders)

3. USER ROLES

SDIS operates with two distinct user roles. Each role has specific capabilities and boundaries — neither can do everything, and that separation is intentional.

DETECTIVE

The primary operational user. Detectives manage the day-to-day investigative work — but they do not start with automatic access to anything. All capabilities must be requested and approved first.

What a Detective can do (once approved):

- Request permission to create or review cases
- Create new cases with evidence and suspects
- Review their own cases and verify evidence integrity
- Modify evidence descriptions (all modifications are logged)

COMMANDER

The supervisory oversight role. Commanders do not create or manage cases directly — their function is to control access and monitor system activity. Nothing moves without their approval.

What a Commander can do:

- Review all pending access requests from Detectives
- Approve or deny those requests individually
- View the complete login history of the system
- View the full evidence modification history (tamper log)

4. HOW ACCESS WORKS

SDIS uses a request-and-approval model. No Detective can simply log in and start working — access to each capability must be explicitly granted by a Commander. This reflects real organizational structures where senior officers control what investigators are permitted to do.

The Access Flow

Step 1 — A Detective logs in with their credentials.

Step 2 — They submit a request for a specific permission: either to create cases, or to review cases. These are separate permissions.

Step 3 — The request enters a pending queue visible only to the Commander.

Step 4 — The Commander reviews the request and either approves or denies it.

Step 5 — If approved, the Detective can now perform that action. If denied, they remain blocked until they request again and a Commander approves.

This separation of powers means that even if a Detective's login credentials were somehow compromised, an attacker still could not act without Commander approval. The two roles are designed to check each other.

5. CASE MANAGEMENT

Cases are the central unit of the system. Each case is a complete record of an investigation, containing all the information a Detective logs during their work.

What a Case Contains

- **Case Number — a unique identifier for the case (e.g. CAS-2026-001)**
- Crime Type — the category of the incident (e.g. Fraud, Theft, Cybercrime)
- Summary — a written description of the incident
- Evidence — one or more pieces of digital or physical evidence, each with a description
- Suspects — named individuals with a reason for suspicion and a suspicion score
- Creator — the username of the Detective who opened the case
- Integrity Hash — a mathematical fingerprint of the case taken at the moment of creation

Cases are private to the Detective who created them. Even if multiple Detectives are using the system simultaneously, each can only review their own cases. This enforces ownership separation and prevents cross-contamination of investigations.

6. EVIDENCE INTEGRITY

The most technically significant feature of SDIS is its evidence integrity system. The moment a piece of evidence is entered into the system, a unique numerical fingerprint — called a hash — is calculated from its description and stored alongside it.

How Integrity Checking Works

When a Detective logs evidence, SDIS calculates a hash of the evidence description and stores it. Every character in the text contributes to the hash number. Change even one letter in the description and the hash changes entirely.

When the case is reviewed later, SDIS recalculates the hash on the spot and compares it to the stored value. If they match, the evidence is flagged as intact. If they differ — even by a single character — the system immediately warns: EVIDENCE TAMPERED.

Case-Level Integrity

This same principle applies to the case itself. When a case is created, a hash of the case number, crime type, and summary is stored. On review, the hash is recalculated and compared. Any unlogged changes to the core case fields are immediately detectable.

> *In real digital forensics, evidence that cannot be proven unaltered may be inadmissible in court. SDIS makes this verification automatic and instantaneous.*

7. EVIDENCE MODIFICATION & AUDIT TRAIL

Detectives are permitted to modify evidence descriptions after the fact. This is a deliberate design choice — investigators may need to correct or expand records as a case develops.

However, every modification is automatically recorded in a tamper log. The system captures: who made the change, which case was affected, what action was taken, and exactly when it happened.

Why the Integrity Hash Is Not Updated on Modification

When evidence is modified, the stored hash is intentionally left as-is — the original fingerprint is preserved. This means that the next time the case is reviewed, the system will detect the discrepancy and display a warning.

This is not a flaw — it is a feature. It creates a detectable paper trail. Even legitimate edits show up as discrepancies, which a Commander can then cross-reference against the tamper log to confirm they were made by an authorized Detective at a logged time. Unauthorized edits have no corresponding log entry, making them immediately suspicious.

8. AUDIT LOGGING

SDIS maintains two independent logs that together provide a complete picture of system activity.

Login History

Every successful login — for both Detectives and Commanders — is recorded with the username, role, and timestamp. The Commander can review this log at any time to see who has been active in the system and when.

Tamper / Modification History

Every time a Detective uses the Modify Evidence function, a record is written to the tamper history. This includes the Detective's username, the case number, the type of action performed, and the exact date and time.

The tamper log is cumulative — it is never cleared during a session. A Commander reviewing this log sees the full history of every modification made since the system started.

9. AUTHENTICATION

SDIS has separate login flows for Detectives and Commanders. To enter the system, a user selects their role and provides a username and password.

- Detectives share a role password to authenticate into the Detective context.
- Commanders have their own separate password to access the Commander context.
- Usernames are individual — each Detective has their own identity within the system, even though they share a password.

- All login attempts — whether successful or not — do not grant automatic access to any capability. Access must be separately requested and approved.

The current implementation uses passwords suitable for a prototype environment. A production deployment would use individual hashed credentials stored in an encrypted, persistent data store.

10. FEATURE REFERENCE

The following table summarizes every capability in the system, who can use it, and how it functions.

FEATURE	ROLE	DESCRIPTION
Login	Both	Password-based authentication. Sets the active user and role for the session.
Request Create Access	Detective	Submits a request to the Commander queue asking for permission to create cases.
Request Review Access	Detective	Submits a request to the Commander queue asking for permission to review cases.
Create Case	Detective (approved)	Builds a full case record: case number, crime type, summary, evidence items, and suspects. Hashes are calculated at creation.
Review Case	Detective (approved)	Displays full case details. Automatically recalculates hashes and compares against originals. Issues warnings on mismatch.
Modify Evidence	Detective	Updates an evidence description. The original hash is preserved. The modification is automatically written to the tamper log.
Approve / Deny Requests	Commander	Reviews the pending request queue and grants or revokes access permissions for individual Detectives.
View Login History	Commander	Displays every login event recorded in the session — username, role, and timestamp.
View Tamper History	Commander	Displays every evidence modification recorded in the session — who changed what, in which case, and when.

11. KEY DESIGN PRINCIPLES

SDIS was built around four principles that reflect how real forensic and law enforcement software should operate.

Role-Based Access Control

Different users have different capabilities. Detectives and Commanders are not interchangeable. This limits what any single user can do and reduces the damage a compromised account can cause.

Request-and-Approval Workflow

Access is not given freely — it must be earned through a formal approval process. This creates accountability and means that capability expansion is always a deliberate, human decision.

Data Integrity Verification

Hash fingerprints make tampering detectable automatically. No manual comparison or external tool is needed. The system flags discrepancies the moment a case is reviewed.

Automatic Audit Logging

Every login and every modification is permanently recorded without any manual step. The audit trail cannot be skipped — it is a built-in consequence of using the system, not an optional feature.

Ownership Enforcement

Detectives can only access cases they created. Investigators work in isolated spaces, preventing cross-contamination of active investigations and unauthorized access to other Detectives' case files.

SECURING EVIDENCE. ENSURING INTEGRITY. DELIVERING JUSTICE.

SDIS — Hackathon Project | 2026 | Built entirely in C++17 Standard Library

Project Video Submission

Please click this link below to watch our video presentation/project pitch:

https://drive.google.com/file/d/1B-mX4U-uHA9Uz5xcgFfu9EpBDtl5Lojx/view?usp=sharing